

See discussions, stats, and author profiles for this publication at: <https://www.researchgate.net/publication/274411304>

Experimental investigation on friction factor in pipes with large roughness

Article in *Experimental Thermal and Fluid Science* · October 2013

DOI: 10.1016/j.expthermflusci.2013.06.002

CITATIONS

64

READS

6,197

6 authors, including:



Kun Huang

China University of Geosciences

30 PUBLICATIONS 437 CITATIONS

SEE PROFILE



Experimental investigation on friction factor in pipes with large roughness

K. Huang^{a,*}, J.W. Wan^{a,*}, C.X. Chen^a, Y.Q. Li^b, D.F. Mao^c, M.Y. Zhang^a

^a School of Environmental Studies, China University of Geosciences, 430074 Wuhan, China

^b Three Gorges Geotechnical Consultants Co., Ltd., Wuhan, China

^c WISDRI Engineering & Research Incorporation Limited, Wuhan, China

ARTICLE INFO

Article history:

Received 6 March 2013

Received in revised form 2 June 2013

Accepted 2 June 2013

Available online 14 June 2013

Keywords:

Friction factor

Laminar flow

Transitional flow

Surface roughness

ABSTRACT

This paper experimentally and theoretically investigated the differences on friction factor in rough pipes with large surface roughness. Deviation from theoretical theory for laminar flow in rough tubes was confirmed again. The product of fRe was larger than 64 and can be predicted as a quadratic equation of relative roughness. Earlier transition between flow regimes was observed. The critical Reynolds number between flow regimes decreased as the relative roughness increased. The range of Reynolds number for transitional flow regime gradually reduces with an increase in surface roughness. When relative roughness reach peak ($\Delta/d = 1/2.4$), transitional flow regime disappeared. Navier–Stokes equation was used to analyze the differences on flow behavior. It indicates flow obstruction of surface roughness caused curve flow for laminar flow in porous medium and linear law for laminar flow in porous media may be an approximate expression by neglecting the inertial forces at low Reynolds number.

© 2013 Elsevier Inc. All rights reserved.

1. Introduction

Moody chart characterized by Reynolds number (Re) and friction factor (f) was always used to analyze pressure drop experimental data during internal flow. The pioneering work on the friction factor were studied by Darcy [1] and Fanning [2] in the nineteenth century, however the systematic work on understanding the relationship of Reynolds number and friction factor was conducted by Nikuradse [3]. According to the experimental results, Nikuradse mainly divided the flow regimes into three portions of ranges:

- (1) Laminar flow: the friction factor decreased linearly as the Reynolds number increased, and the friction factor can be expressed as:

$$fRe = 64 \quad (1)$$

Where

$$f = \frac{2gd}{u^2} J \quad (2)$$

in which J is the hydraulic gradient means the head drop referred to a length unit of the tube, d the diameter of the test tube, u is the average flow velocity and

$$Re = \frac{ud}{\nu} \quad (3)$$

in which ν is the kinematic viscosity. We can know that the hydraulic gradient is proportional to the velocity in laminar flow ($J \propto u$) from Eq. (1).

- (2) Transitional flow: the friction factor increased as the Reynolds number increased, and the relationship between hydraulic gradient and velocity can be written as follows:

$$J = u^m \quad (4)$$

where the value of m range from 1 to 2 [4].

- (3) Turbulent flow: the friction factor independent of the Reynolds number and the hydraulic gradient can be predicted by the formula:

$$J = u^2 \quad (5)$$

Thus Eq. (5) was also called the quadratic law of resistance. The conclusion of Nikuradse' experiments were the classical theory on flow in rough pipes.

The development on the theory of flow in porous media was partly inspired by the research methods of flow in conduits. The first law on flow in porous media was proposed by Darcy

* Corresponding authors.

E-mail addresses: cugdr_huang@163.com (K. Huang), wanjw@cug.edu.cn (J.W. Wan).

Nomenclature

A	cross-section, m^2
d	diameter, m
Δ	surface roughness, m
Re	Reynolds number
f	friction factor
u	mean velocity, m s^{-1}
J	hydraulic gradient
ΔH	head loss, m
P	pressure, Pa
Q	flow rate, $\text{m}^3 \text{s}^{-1}$

Greek symbols

ν	kinematic viscosity, $\text{m}^2 \text{s}^{-1}$
μ	dynamic viscosity, $\text{kg m}^{-1} \text{s}^{-1}$
ρ	liquid density, kg m^{-3}

Subscripts

p	porous media
cf	constricted flow
l	laminar flow
t	transitional flow

conducting experiments through filter sands [1]. The expression of Darcy's law that shows a linear relationship between hydraulic gradient and velocity is shown in Eq. (6) as follows:

$$u = KJ \quad (6)$$

in which K a factor of proportionality. In a long time, Darcy's law was the only equation describing flow in porous media until deviation from Darcy's law was firstly observed by Forchheimer at high velocity [5,6]. Forchheimer presented a non-linear equation with a quadratic term for high velocity flow [7] which is shown in Eq. (7) as follows:

$$J = au + bu^2 \quad (7)$$

where a and b are constants of porous media and fluid. The quadratic law was then confirmed by many experimental data [8–10]. Hence, many researchers investigated the critical point between flow regimes in porous media in order to better understand the validity of the Darcy's law [11,12]. Due to previous belief that flow in porous media was similar to flow in conduits, Reynolds number and moody chart were used to determine the deviation from linear flow to non-linear flow [13]. Reynolds number in porous media was defined as:

$$Re = \frac{ud_p}{\nu} \quad (8)$$

where d_p was the characteristic length of particles. The early systematic work on understanding the relationship of Reynolds number and friction factor in porous media was conducted by Arbabhirama and Dinoy [15]. They analyzed experimental data from previous investigations and proposed the empirical relationship between Reynolds number and friction factor in porous media (Fig. 2). It showed that there are four kinds of flow regimes in

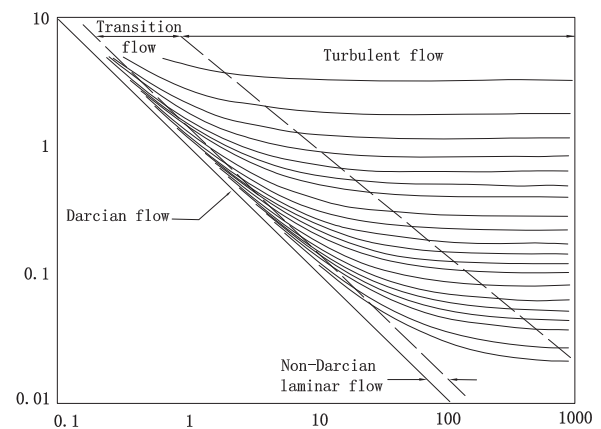


Fig. 2. Flow regimes in porous media (Arbabhirama and Dinoy [15]).

porous media: Darcian flow, non-Darcian laminar flow, transition flow and turbulent flow.

Comparing the two figures (Figs. 1 and 2), we can easily found that the characteristic of the moody chart for porous media is quite different from rough pipes. According to the experimental data from previous studies [14,15], the relationship curve for Reynolds number versus friction factor is smooth, and there was no sharp limit between laminar flow and transition flow. The critical Reynolds number obtained by previous researchers could arrange from 1 to 1000 for different kind of porous media [16,17]. Until now, there still was no widely accepted criterion for non-linear flow in porous media [12,18,19], but the critical point between laminar flow and transition flow is quite steady with the Reynolds number around 2000 [3,20]. However, the problem that differences of flow characters between porous media and conduits attracted not enough attention by investigators.

In recently years, experimental results in micro scale level have found some new characters in laminar flow. Earlier transition from

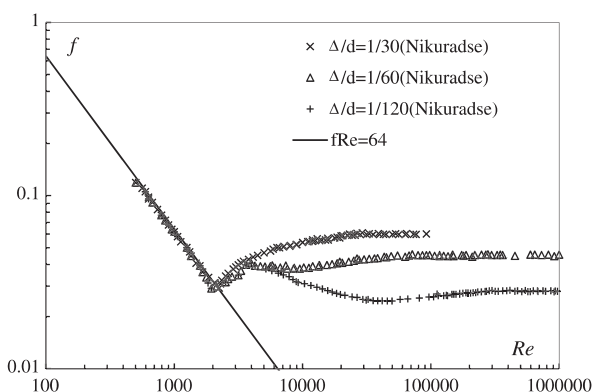


Fig. 1. Nikuradse's experimental data in rough pipes.

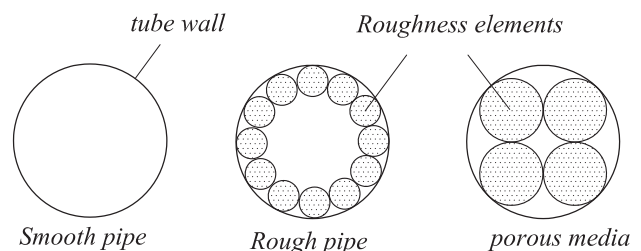


Fig. 3. Schematic of design scheme for test tubes.

laminar flow to transition flow has been observed by several investigators [21–23]. The earlier transition was also confirmed by our experiments on flow in pipes with large relative roughness than Nikuradse' [24]. Another interesting phenomenon that the region of transition flow decreased as the relative roughness increased was also observed. Inspired by the above experimental results, we can design a series of experiments to investigate flow in pipes with larger roughness. If the surface roughness reaches the peak when particles used to create surface roughness is so large that particles contact with each other, then the rough pipe turns into porous media (Fig. 3). Based on the above consideration, experiments on flow in pipes with large relative roughness and porous media were both conducted. Further, the experimental results on flow in the two medium were compared with each other. Possible reason for the phenomenon of the differences between flow behaviors in rough pipes and porous media was proposed.

2. Experimental setup and data reduction

2.1. Experimental setup

The experimental apparatus used in this investigation is depicted schematically in Fig. 4. It mainly consists of three parts: an adjustment valve, test section (rough tubes) and measurement equipment. Water was supplied by the centrifugal pump to the test tube and flow rate was adjusted by the inflow valve. After passing through the test tube, water was collected by a tank. The fluid flow rate was calculated by water volume collected over a corresponding measured elapsed time. To minimize errors, flow rate were observed three times for a certain head drop and mean value was adopted. Three piezometer tubes were used to measure the water head drop. The location of the piezometer tubes was shown in Fig. 4. Piezometer tubes were 500 mm away from both ends of the test tube to decrease the inlet and outlet effects. Meanwhile, water temperature was also measured by thermometer.

In our earlier study, we have conducted experiments in six organic glass tubes with different relative roughness (Δ/d) to investigate the deviation from classical theory on laminar flow in pipes with large roughness [24]. The surface roughness was created by sticking sifted sands to the inside walls of the tubes. The information of rough tubes with different relative roughness is shown in Table 1. In this study, another five tubes with high surface roughness were made. Similar method was adapted to create roughness. Considering that morphological difference will increase as the diameter of the sands increase, acrylic spheres with the same diameter were chosen instead of sifted sands to keep uniform roughness throughout the tubes. As the same with the earlier research, chloroform was used to stick the spheres to the walls of

Table 1

Relative roughness results in former research.

Number	d (mm)	Δ (mm)	Sieves diameter (mm)	Δ/d
R1	19	0.16	0.15–0.17	1/118.75
R2		0.3075	0.3–0.315	1/61.79
R3		0.615	0.6–0.63	1/30.89
R4		1.34	1.25–1.43	1/14.18
R5		1.8	1.6–2	1/10.56
R6		2.85	2.5–3.2	1/6.66

Table 2

Relative roughness results in this study.

Number	d (mm)	Δ (mm)	Δ/d
R7	19	3.91	1/4.86
R8	19	4.5	1/4.22
R9	19	5.25	1/3.62
R10	19	6.33	1/3
R11	24	10	1/2.4

the tubes and the refilled process in Nikuradse' experiments was not adapted. Each test tube is 3000 mm in length. The test tubes with surface roughness (Δ/d) equal to 1/4.86, 1/4.22, 1/3.62 and 1/3 were 19 mm in inner diameter (d). Since there is no standard size acrylic spheres adapting to the tube 19 mm in inner diameter, acrylic spheres 10 mm in diameter and tube 24 mm in inner diameter were chosen to make test tube with peak relative roughness equal to 1/2.4. The information of rough tubes with different relative roughness for this study is shown in Table 2.

2.2. Data reduction

According to the experimental data, hydraulic gradient and average flow velocity can be calculated by following formula:

$$J = \frac{\Delta H}{\Delta L} \quad (9)$$

where ΔH is the water drop between piezometer tubes and ΔL is the length between piezometer tubes.

$$u = Q \cdot \frac{4}{\pi d^2} \quad (10)$$

where u is the average flow velocity, Q is the measured flow rate and d is the diameter of the tube.

Reynolds number (Re) and friction factor (f) were calculated by hydraulic gradient and average flow velocity. In order to compare experimental results to the former studies, Re and f were defined

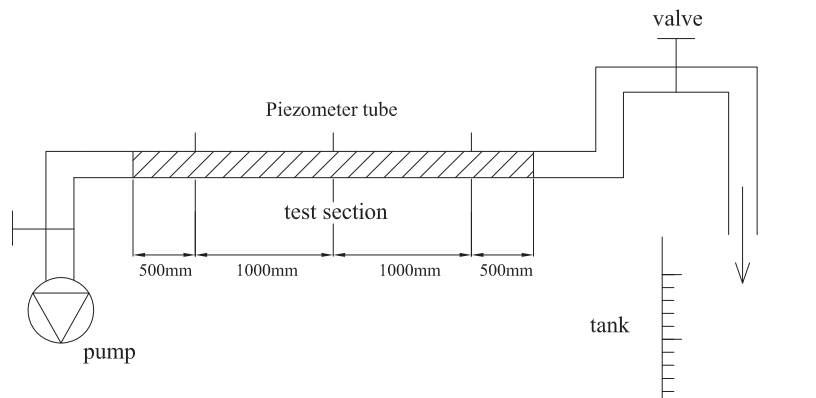


Fig. 4. Schematic of experimental setup.

Table 3

Experimental uncertainties of the major parameters.

Parameter	Uncertainty	Parameter	Uncertainty
Diameter (d)	$\pm 1\%$	Time (t)	$\pm 0.1\text{s}$
Temperature (T)	$\pm 0.1\text{ }^\circ\text{C}$	Volume (V)	$\pm 1\%$
Head (H)	$\pm 1\text{ mm}$	Reynolds number (Re)	$\pm 3.2\%$
Lengths (L)	$\pm 1\text{ mm}$	Friction factor (f)	$\pm 5.9\%$

as the same with Nikuradse. Reynolds number was calculated from equation with the following form:

$$Re = \frac{ud}{\nu} \quad (11)$$

in which ν is the kinematic viscosity. The friction factor f is determined by Eq. (2).

The experimental uncertainties of the parameters were listed in Table 3. The uncertainty of the Reynolds number and the friction factor is determined by equation using the method described by Abernethy [25].

$$y = f(x_1, x_2, \dots, x_N)$$

$$\delta y = \sqrt{\left[\left(\frac{\partial f}{\partial x_1} \delta x_1 \right)^2 + \dots + \left(\frac{\partial f}{\partial x_N} \delta x_N \right)^2 \right]} \quad (12)$$

By applying Eq. (12), the calculated mean uncertainty for the Reynolds number and friction factor are 3.2% and 5.9%, respectively.

3. Results and discussion

3.1. Experimental results

Water head drop was measured with the five test tubes at various flow rates. Hydraulic gradient (J) is calculated by Eq. (9). Hydraulic gradient (J) was plotted as a function of flow velocity for the five rough tubes with relative roughness $\Delta/d = 1/4.86$, $1/4.22$, $1/3.62$ and $1/3$ and $1/2.4$ (Fig. 5). It shows that the value of J increased as the value of Re increased for all tubes. With the same flow velocity, value of J is greater for tubes with larger relative roughness. It indicated that larger roughness caused higher friction and water drop. Accordingly, it would get a higher flow velocity in tubes with lower surface roughness under the same hydraulic gradient.

To investigate the effects of surface roughness on flow in rough tubes, Moody chart was chosen to plot the experimental data. To find out the differences of flow characteristic between flow in

tubes and porous media, experimental results of tubes with lower surface roughness were quoted to compare with experimental data in this study (Fig. 6). We can see from Fig. 6 that the experimental curves can divide into three kinds as follows:

3.1.1. Flow in rough tubes with low surface roughness

The experimental data for tubes with relative roughness $\Delta/d = 1/118.75$, $1/61.79$, $1/30.89$ (R1, R2, R3) were in good agreement with the classic theory. The friction factor was not affected by relative roughness in laminar flow and the value of fRe is equal to 64. It can also be apply to flow in microchannels which has been confirmed by Morini et al. on a discussion about the lack of influence of the relative roughness on friction factor [26]. The deviation from laminar to turbulent flow occurred around $Re = 2000$ in accord with Nikuradse' experiments.

3.1.2. Flow in rough tubes with high surface roughness

The experimental data for tubes with relative roughness ranging from $1/14.18$ to $1/3$ (R4–R10) deviated from the conditional theory. The most obvious feature is that the friction factor depends on not only the Reynolds number but also the relative roughness. The product of fRe in laminar flow is no longer equal to 64 but increased with an increasing relative roughness. The relationship between relative roughness and product of fRe was plotted in Fig. 7. It shows that product of fRe was a quadratic function of relative roughness. The correlation of experimental data gave the following equation with $R^2 = 0.9927$. When the relative roughness is small enough (less than the thickness of boundary layer), the value of is nearly equal to 64, while surface roughness caused a sharp increase of fRe for laminar flow when the relative roughness is larger than 5%. Similar conclusion was also presented by other researchers such as Kandlikar [27] and Brackbill [23].

Secondly, earlier transition between flow regimes was confirmed again by experiments. The critical Reynolds number Re_t which is the value of Re for which the friction factor increases with respect to the previous value means the value of Reynolds number at the point from laminar flow to transition flow. The value of Re_t was plotted as a function of relative roughness (Fig. 8). It can be seen from Fig. 8 that the critical Reynolds number Re_t decreased linearly as the relative roughness increased. In other words, the range of laminar flow regime gradually reduced. The critical Reynolds number has been modeled as a line fit and is given as $Re_t = 2304.47 - 6340.3(\Delta/d)$ with a coefficient of determination (R^2) of 0.9945. Similar linear correlation with higher slope was proposed by Brackbill and Kandlikar to predict the critical Reynolds number [28]. The correlation to predict the critical Reynolds number for flow in macro tubes still need further studies.

Earlier transition from transition flow to turbulent flow was observed as well. Critical points between transitional flow and turbulent flow was determined by critical Reynolds number Re_t . Fig. 9 shows the variation of the critical Reynolds number Re_t with relative roughness. As seen from the figure, the value of Re_t falls exponentially with an increase in relative roughness. It indicated that the value of Re_t decreases more sharply than Re_t . So the range of Reynolds number of transitional flow regime is smaller in a tube with larger surface roughness (Fig. 6).

3.1.3. Flow in simple porous media

The tube with peak relative roughness equal to $1/2.4$ represents the simplest ideal model of porous media. We can see from Fig. 6, there is no abrupt transition between flow regimes for flow in tube

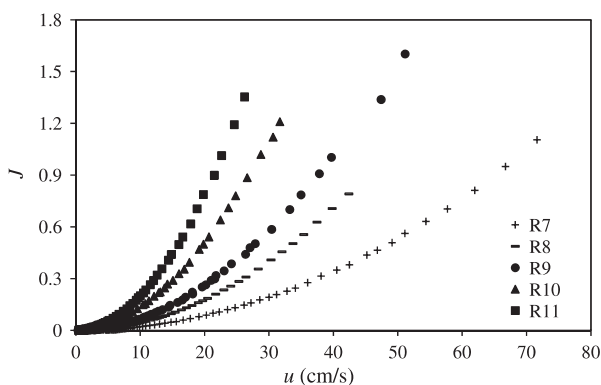


Fig. 5. Variation of hydraulic gradient J with Reynolds number Re .

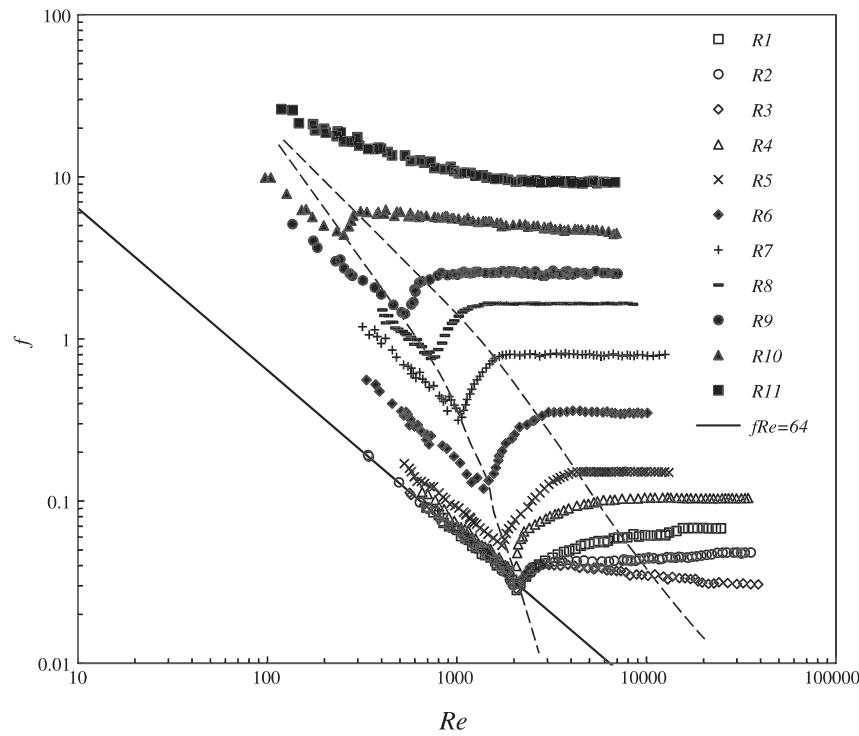


Fig. 6. Variation of friction factor f with Reynolds number Re .

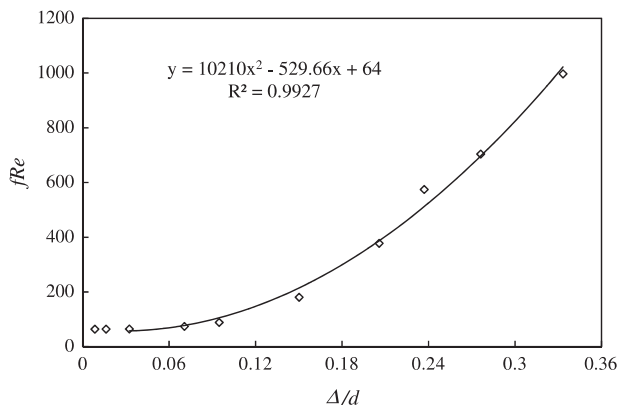


Fig. 7. Product of fRe vs. relative roughness Δ/d .

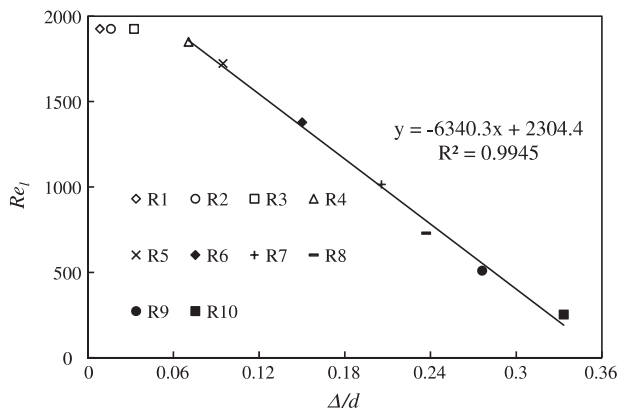


Fig. 8. Critical Reynolds number Re_c vs. relative roughness Δ/d .

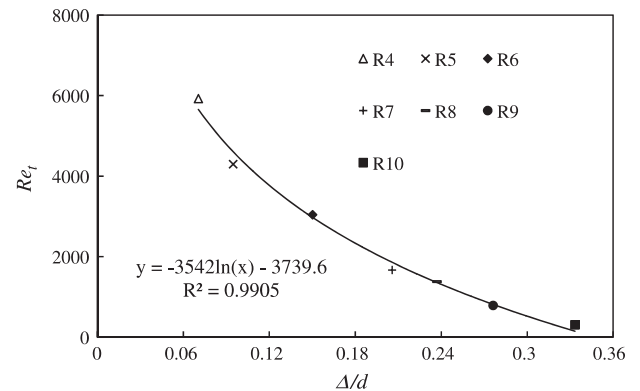


Fig. 9. Critical Reynolds number Re_c vs. relative roughness Δ/d .

R11. The moody chart for porous media is a smooth curve from laminar flow to turbulent flow.

Comparing the experimental results from tube R1 to R11, we obviously observed that both the range of laminar flow and transitional flow regimes tends to reduce as the surface roughness goes up and the transitional flow seems disappear in porous media (R11) when the surface roughness of the tube reach peak. The results of experiments conducted in cubic arrays of spheres also confirmed that there was no significant limits between the flow regimes [29]. Laminar flow in porous media is nonlinear even at low Reynolds number which is quite different from flow in conduits. It will be discussed in the next part.

3.2. Theoretical analysis

Water flow in rough tube in this study can be considered as one-dimensional steady flow in the x direction through the length axis of the tube. Flow viscous incompressible fluid obeys the Navier–Stokes equation. At any point in the tube the flow equation can be written as:

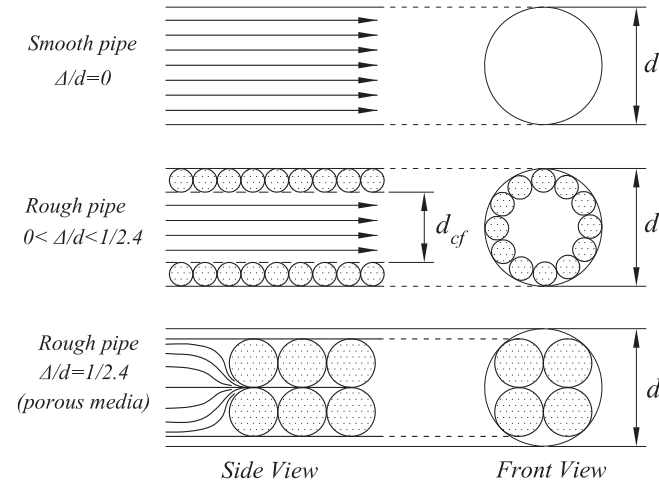


Fig. 10. Flow pattern in rough pipes.

$$\frac{\partial P}{\partial x} = \mu \left(\frac{\partial^2 u_x}{\partial x^2} + \frac{\partial^2 u_x}{\partial y^2} + \frac{\partial^2 u_x}{\partial z^2} \right) - \rho \left(u_x \frac{\partial u_x}{\partial x} + u_y \frac{\partial u_x}{\partial y} + u_z \frac{\partial u_x}{\partial z} \right) \quad (13)$$

where x, y, z were the axes of rectangular coordinate system; u_x, u_y, u_z were the corresponding components of velocity and u_x was the flow velocity in the direction along the length axis of the tube. The first term in the right of Eq. (13) is the viscous term which is proportional to the flow velocity. The second term in the right of Eq. (13) is the inertial term which is the second-order term of the flow velocity.

For laminar flow in the smooth tube, the flow lines were parallel along the tube (Fig. 10). Hence,

$$u_y = 0; u_z = 0$$

For steady flow, there was no acceleration of fluids in the flow direction

$$\frac{\partial u_x}{\partial x} = 0$$

So

$$\left(u_x \frac{\partial u_x}{\partial x} + u_y \frac{\partial u_x}{\partial y} + u_z \frac{\partial u_x}{\partial z} \right) = 0$$

Thus, laminar flow in smooth conduits is linear flow with no inertial term and the friction factor formula can be given by theoretical derivation as following:

$$f = \frac{64}{Re} \quad (14)$$

For laminar flow in rough tubes, deviation from the traditional theory can be mainly attributed to the following two reasons. Firstly, the roughness elements reduced the actual diameter of flow. The effective diameter was called constricted flow diameter (d_{cf}) by Kandlikar, and it was defined as:

$$d_{cf} = d - 2\Delta \quad (15)$$

where d_{cf} is the constricted flow diameter, d is the inner diameter of the tube, and Δ is the surface roughness. Thus, the diameter used to calculate the Reynolds number and friction factor should be replaced by constricted flow diameter. The Reynolds number and friction factor were redefined as follows:

$$Re_{cf} = \frac{u_{cf} d_{cf}}{\nu} = \frac{d}{d_{cf}} \cdot Re \quad (16)$$

$$f_{cf} = \frac{2g d_{cf}}{u^2} \cdot J = f \cdot \frac{d_{cf}^5}{d^5} \quad (17)$$

So the product of fRe can be written as follows:

$$fRe = 64 \cdot \frac{d^4}{(d - 2\Delta)^4} \quad (18)$$

For laminar flow in smooth pipe ($\Delta = 0$), Eq. (18) turned into Eq. (14) where the value of fRe is equal to 64. It indicated that the product of fRe would increase as the surface roughness increases. When the surface roughness is small enough ($\Delta/d < 1/30$), the effects of surface roughness on laminar flow can be neglected. But for tubes with larger surface roughness, deviation from classic theory gradually becomes obvious. The above analysis has been verified by the experimental results. However, Eq. (18) could not be directly applied to predict the product of fRe for rough pipes because another effect of surface roughness on flow is flow obstruction [30]. Obstructed by the roughness elements, flow lines near the roughness elements would become curves instead of parallel lines. For example, it was observed that flow separated from the surface of the roughness elements and then reattached at a distance 6–8 times the height of the elements [31]. Thus, for laminar flow in rough tubes, the flow pattern consists of two parts: parallel linear flow in the constricted area and curve flow near the roughness elements. We can know from Eq. (13), the curvature of flow lines would cause component velocities in y and z direction and the inertial term in the equation is not equal to zero. As we explained above, inertial term causes the nonlinear behavior which is the second-order term of the flow velocity. The value of inertial term is much smaller than viscous term which can be neglected when flow velocity is very low. As the velocity reach a certain value, the nonlinear behavior becomes significant. For laminar flow in tubes with lower surface roughness, the proportion of curve flow in the tube was smaller. Therefore, it explained why the transition from laminar flow to transitional flow should be earlier in the tubes with higher surface roughness.

However, for laminar flow in rough tube with peak surface roughness or porous media, the curvature and variable cross-section of flow channels would cause frequent changes and acceleration in flow direction. There is no parallel linear flow but curve flow. It indicates that flow in porous media should be nonlinear flow even at low velocity. Similar conclusion was also presented by other investigators who believed that non-linear behavior was attributed to the inertial forces which were caused by frequent changes in flow direction and the acceleration of fluids passing through the curved channels in porous media [6,32,33]. Consequently, the linear law for laminar flow in porous media may be an approximate expression by neglecting the inertial forces at low Reynolds number. This deduction will be discussed elsewhere by experimental investigation.

4. Summary and conclusions

We obtained a series of experimental data investigating the one-dimensional steady flow in rough tubes. The main conclusions can be summarized as follows:

- Deviation from theoretical theory for laminar flow in rough tubes was confirmed again by experiments; the product of fRe was a quadratic function of relative roughness which can be predicted by equation when relative roughness $\Delta/d > 1/30$: $fRe = 10210(\Delta/d)^2 - 529.66(\Delta/d) + 64$.
- Earlier transition between flow regimes was observed; the critical Reynolds number Re_t decreased linearly as the relative roughness increased and Re_t falls exponentially with an increase in relative roughness.

- The range of Reynolds number for transitional flow regime gradually reduces with an increase in surface roughness; for flow in porous media with the peak surface roughness, transitional flow regime disappeared.
- Different characteristic on friction factor between flow in rough tubes and porous medium was due to the differences of flow pattern; flow obstruction of surface roughness caused curve flow for laminar flow in porous medium; it indicates that the linear law for laminar flow in porous media may be an approximate expression by neglecting the inertial forces at low Reynolds number.

Acknowledgements

This study was supported by the National Natural Science Foundation of China (Grant No. 40972155) and the Fundamental Research Funds for the Central Universities (CUG120817), China University of Geosciences (Wuhan). The authors would like to thank for their financial support. Thank engineer Shen Zhongzhi of China University of geosciences (Wuhan) for great help in developing experimental set up.

References

- [1] H. Darcy, Recherches expérimentales relatives au mouvement de l'eau dans les tuyaux, Mallet-Bachelier, 1857.
- [2] J.T. Fanning, A Practical Treatise on Hydraulic and Water-Supply Engineering: Relating to The Hydrology, Hydrodynamics, and Practical Construction of Water-Works, D. Van Nostrand Company, North America, 1906.
- [3] J. Nikuradse, Laws of Flow in Rough Pipes, National Advisory Committee for Aeronautics, Washington, 1950.
- [4] O. Reynolds, An experimental investigation of the circumstances which determine whether the motion of water shall be direct or sinuous, and of the law of resistance in parallel channels, Proc. Royal Soc. London 35 (1883) 84–99.
- [5] J. Bear, Dynamics of Fluids in Porous media, Dover publications, 1988.
- [6] K.N. Moutsopoulos, I.N.E. Papaspyros, V.A. Tsihrintzis, Experimental investigation of inertial flow processes in porous media, J. Hydrol. 374 (2009) 242–254.
- [7] P. Forchheimer, Wasserbewegung durch boden, Z. Ver. Deutsch. Ing. 45 (1901) 1782–1788.
- [8] S. Ergun, A.A. Orning, Fluid flow through randomly packed columns and fluidized beds, Ind. Eng. Chem. 41 (1949) 1179–1184.
- [9] C. Chukwudozie, M. Tyagi, S. Sears, C. White, Prediction of non-darcy coefficients for inertial flows through the castlegate sandstone using image-based modeling, Trans. Porous Media 95 (2012) 563–580.
- [10] O.O. Alabi, Validity of Darcy's law in laminar regime, Electron. J. Geotech. Eng. 16 (2011) 27–40.
- [11] P. Venkataraman, P. Rama Mohan Rao, Darcian, transitional, and turbulent flow through porous media, J. Hydraulic Eng. 124 (1998) 840–846.
- [12] Z. Zeng, R. Grigg, A criterion for non-Darcy flow in porous media, Trans. Porous Media 63 (2006) 57–69.
- [13] T.H. Chilton, A.P. Colburn, Pressure drop in packed tubes, Ind. Eng. Chem. 23 (1931) 913–919.
- [14] N. Ahmed, D.K. Sunada, Nonlinear flow in porous media, J. Hydr. Div. 95 (1969) 1847–1857.
- [15] A. Arbhahirama, A.A. Dinoy, Friction factor and Reynolds number in porous media flow, J. Hydraul. Div., Am. Soc. Civ. Eng. (United States) 99 (1973).
- [16] J.P. Du Plessis, J.H. Masliyah, Mathematical modeling of flow through consolidated isotropic porous media, Transp. Porous Media 3 (1988) 145–161.
- [17] G.H. Fancher, J.A. Lewis, Flow of simple fluids through porous materials, Ind. Eng. Chem. 25 (1933) 1139–1147.
- [18] Z.X. Li, Experimental study on flow characteristics of liquid in circular microtubes, Microscale Thermophys. Eng. 7 (2003) 253–265.
- [19] D. Hlushkou, U. Tallarek, Transition from creeping via viscous-inertial to turbulent flow in fixed beds, J. Chromatogr. A 1126 (2006) 70–85.
- [20] M. Lorenzini, G.L. Morini, T. Henning, J. Brandner, Uncertainty assessment in friction factor measurements as a tool to design experimental set-ups, Int. J. Therm. Sci. 48 (2009) 282–289.
- [21] G.M. Mala, D.Q. Li, Flow characteristics of water in microtubes, Int. J. Heat Fluid 20 (1999) 142–148.
- [22] G.H. Tang, Z. Li, Y.L. He, W.Q. Tao, Experimental study of compressibility, roughness and rarefaction influences on microchannel flow, Int. J. Heat Mass Trans. 50 (2007) 2282–2295.
- [23] T.P. Brackbill, S.G. Kandlikar, Effect of triangular roughness elements on pressure drop and laminar-turbulent transition in microchannels and minichannels, in: Proceedings of ICNMM2006 Fourth International Conference on Nanochannels, Microchannels and Minichannels, Limerick, Ireland, 2006.
- [24] K. Huang, J. Wan, C. Chen, D. Mao, Y. Li, Experiments investigation of the effects of surface roughness on laminar flow in macro tubes, Exp. Thermal Fluid Sci. 45 (2013) 243–248.
- [25] R. Abernethy, R. Benedict, R. Dowdell, ASME measurement uncertainty, J. Fluids Eng. 107 (1985) 161–163.
- [26] G. Morini, M. Lorenzini, S. Salvigni, G. Celata, Experimental analysis of microconvective heat transfer in the laminar and transitional regions, Exp. Heat Transfer. 23 (2009) 73–93.
- [27] S.G. Kandlikar, D. Schmitt, A.L. Carrano, J.B. Taylor, Characterization of surface roughness effects on pressure drop in single-phase flow in minichannels, Phys. Fluids 17 (2005) 100606.
- [28] T.P. Brackbill, S.G. Kandlikar, Application of lubrication theory and study of roughness pitch during laminar, transition, and low reynolds number turbulent flow at microscale, Heat Transfer Eng. 31 (2010) 635–645.
- [29] K. Huang, J. Wan, C. Chen, L. He, W. Mei, M. Zhang, Experimental investigation on water flow in cubic arrays of spheres, J. Hydrol. 492 (2013) 61–68.
- [30] S. Kandlikar, Roughness effects at microscale-reassessing Nikuradse's experiments on liquid flow in rough tubes, Tech. Sci. 53 (2005) 343–349.
- [31] R. Webb, E. Eckert, R. Goldstein, Heat transfer and friction in tubes with repeated-rib roughness, Int. J. Heat Mass Trans. 14 (1971) 601–617.
- [32] R. Pietlicki, I.R. Barden, Inertial model of flow through porous media, IAHS Publ.–Series Proc.Rep.– Assoc Hydrol. Sci. 237 (1996) 443–452.
- [33] J.G.I. Hellstrom, P.J.P. Jonsson, T.S. Lundstrom, Laminar and turbulent flow through an array of cylinders, J. Porous Media 13 (2010) 1073–1085.